Introduction

Retinol-binding protein (RBP3) is a major component of the interphotoreceptor matrix (IPM) and supports retinoid transport between the retinal pigment epithelium (RPE) and the photoreceptor cells. Its presence within the interphotoreceptor matrix is light- and dark-dependent. Reduced RBP3 levels have been associated with the progression of diabetic retinopathy, making this protein an attractive candidate biomarker for early diabetic retinopathy detection.

We aim to develop a diagnostic approach in which we can measure the level of RBP3 using fluorescently labelled antibodies combined with two-photon excitation fluorescence (TPEF) imaging. The current diagnostic approach to diabetic retinopathy involves observing structural changes in the retina, which detects the disease at a late stage. Our innovative approach would potentially enable the diagnosis of diabetic retinopathy earlier, giving patients a better chance of maintaining their quality of vision for longer.

Methods

Microinjections of retinas at early and progressive diabetic retinopathy stages, with fluorescently labelled antibodies against RPB3, will be followed by in vivo two-photon imaging. Signal distribution and intensity will be validated ex vivo by immunofluorescence microscopy, Western blot, and mass spectrometry.

Results

Preliminary observation shows RBP3 displays a distinct light-dependent distribution, and is aligned with the changes in IPM volume due to hydration levels. Our preliminary data confirmed a strong concentration of RBP3 at the IPM’s borders in light conditions (the interface with the RPE and the junction of inner and outer segments of photoreceptors. Immunostainings will also show if there are any changes in the pattern of expression of RBP3 after forming complexes with fluorescent tags. The spotlight tool in our experiment is two-photon imaging – only minimally invasive while giving an early picture of starting the diabetic retinopathy.

Aims

Our project aims to develop a minimal invasive platform for tracking RBP3 abundance and distribution in retinal tissue. Additionally, RBP3 could be the therapeutic molecule alleviating the symptoms of diabetic retinopathy, and our proposed detection method can support follow-up checks on the treatments.